Assignment - javascript

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ASSIGNMENT QUESTIONS:

Assignment Title: Deep Dive into JavaScript Fundamentals

1. Understanding Asynchronous JavaScript and the Event Loop:
   Explain in detail how JavaScript, despite being single-threaded, handles asynchronous operations. Describe the components of the JavaScript Event Loop (Call Stack, Web APIs, Callback Queue/Task Queue, Microtask Queue, and the Event Loop itself) and their interaction. Provide a clear code example demonstrating an asynchronous operation (e.g., `setTimeout`, `fetch` or promises) and trace its execution flow through these components, explaining why and when specific callbacks are executed.

2. Exploring Closures and Their Practical Applications:
   Define what a closure is in JavaScript. Explain the mechanism by which closures are created, what they "close over," and their key characteristics. Illustrate your explanation with at least two distinct code examples. For each example, describe a practical use case where closures provide a beneficial solution (e.g., maintaining private state, creating function factories, currying, module patterns).

3. `this` Keyword: Context and Binding Rules:
   The `this` keyword in JavaScript is notorious for its context-dependent behavior. Explain the five main binding rules or contexts that determine the value of `this` (global binding, implicit binding, explicit binding with `call`/`apply`/`bind`, `new` binding, and arrow function lexical binding). Provide a unique code example for each rule, clearly demonstrating how `this` is bound in that specific scenario and explaining the underlying reason.

4. Prototypal Inheritance vs. ES6 Classes:
   JavaScript is fundamentally a prototypal language. Explain the concept of prototypal inheritance, detailing how objects inherit properties and methods from their prototypes. Then, discuss the `class` syntax introduced in ES6. Explain how ES6 classes are syntactic sugar over JavaScript's existing prototypal inheritance model. Provide code examples demonstrating both pure prototypal inheritance and class-based inheritance, highlighting the similarities and differences in their underlying mechanisms and usage.

5. JavaScript Module Systems: CommonJS vs. ES Modules:
   Compare and contrast the two prevalent module systems in JavaScript: CommonJS and ES Modules (ESM). Discuss their syntax (`require`/`module.exports` vs. `import`/`export`), how they resolve dependencies, and their typical use cases (e.g., Node.js vs. browser environments).

Explain the advantages and disadvantages of each system and discuss the complexities and considerations developers face when migrating a project from CommonJS to ESM.
